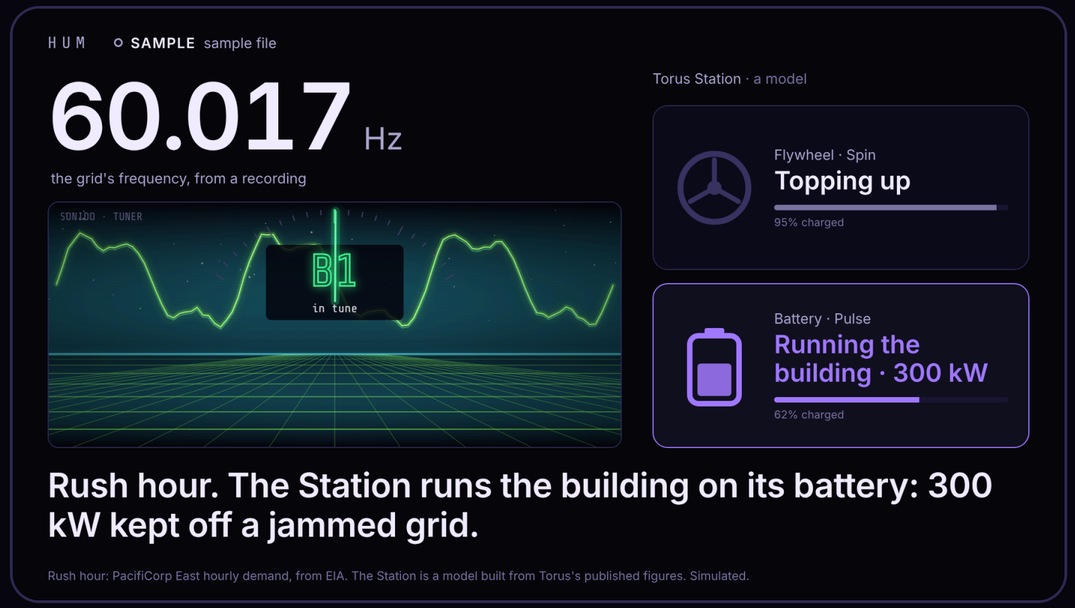
# HUM: the power grid, heard through a guitar, and a Torus Station deciding what to do about it.

Every wire in a building hums at the grid's frequency. That one number is shared by the whole Western grid, from Alberta to Baja.

A guitar pickup hears it. My audio library, sonido, reads it to a thousandth of a hertz and shows it on its tuner.

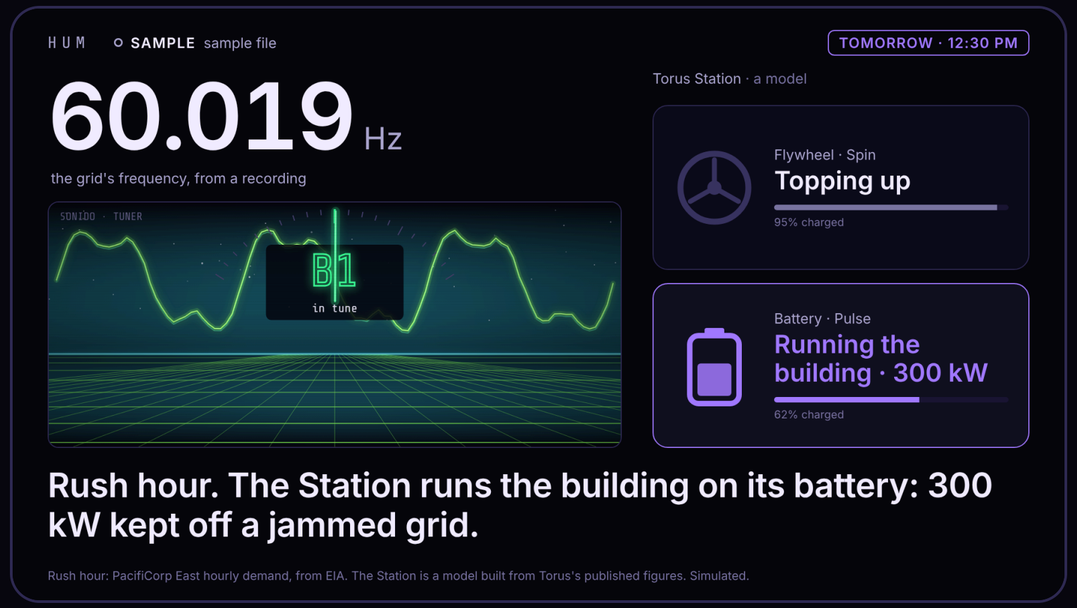


Rush hour you can see coming

# The battery carries the building through the jam.

EIA's hourly data for PacifiCorp East shows when the grid gets jammed: this week, from noon to four.

At tomorrow's noon the Station runs the building on its battery, **300 kW kept off the grid**, and recharges when the power is clean.



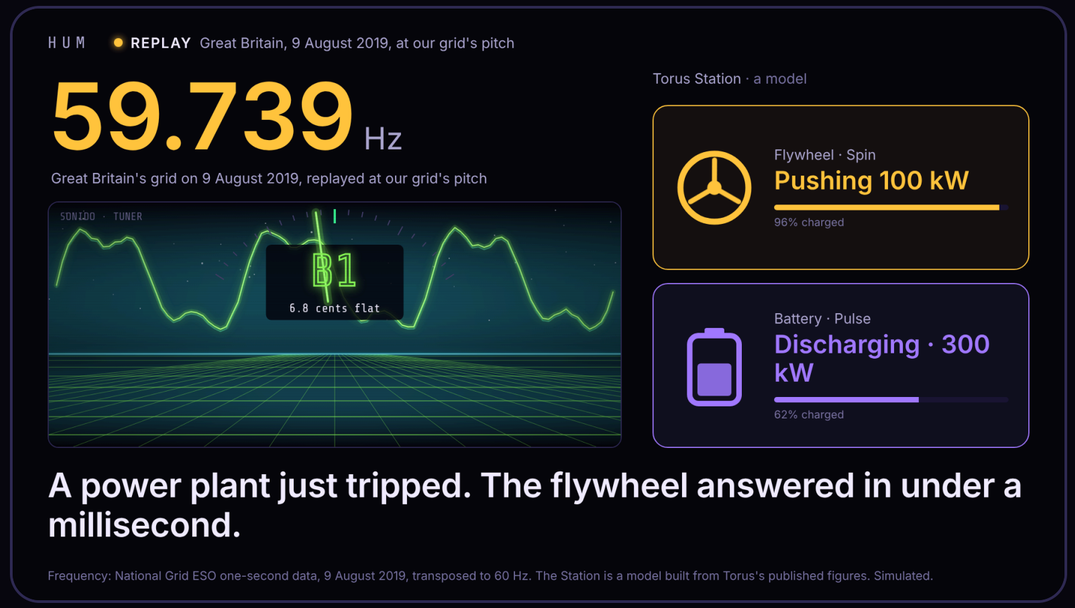

Data: EIA-930, PacifiCorp East. The Station is a model built from Torus's published figures.

A crash you can only hear

# The flywheel answers in under a millisecond.

On 9 August 2019, two power plants in Britain tripped within a second and a million people lost power. Its real frequency record, replayed at our grid's pitch, pulls the number down.

The flywheel pushes at full power before an hourly feed could notice. **When the grid goes dark, the building stays on.**



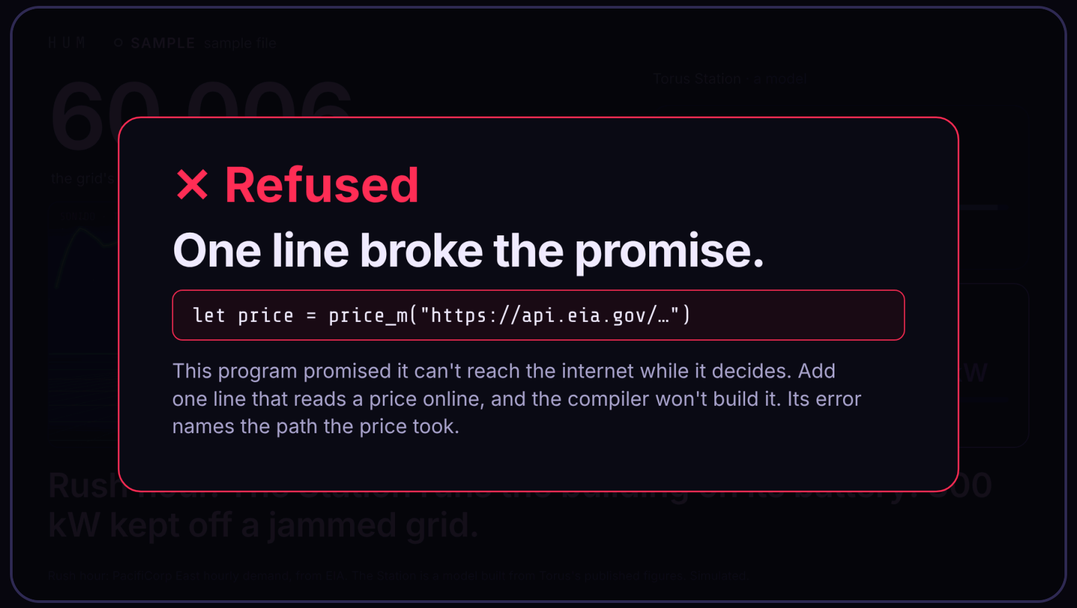

Data: National Grid ESO one-second frequency, 9 August 2019, transposed to 60 Hz.

For your engineers

# The software making these decisions can't reach the internet.

It is written in Mentl, a language I built whose compiler proves that promise through every function the decision calls.

Add one line that reads a price online, and the compiler refuses to build it.




Replay the demo:
`ampactor.dev/hum`
sonofagoose@pm.me
github.com/ampactor-labs/hum

Thank you, Torus.